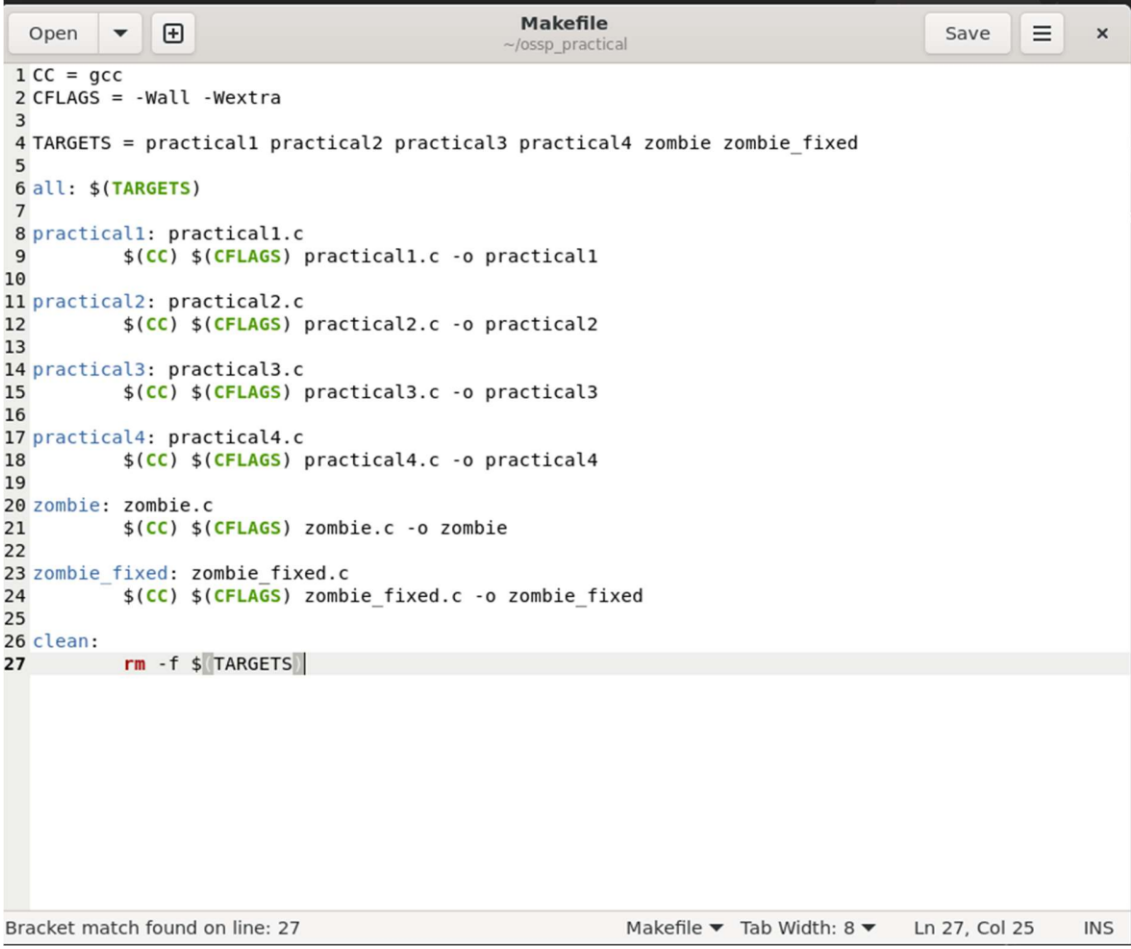


Skill -1 OS



```
1 CC = gcc
2 CFLAGS = -Wall -Wextra
3
4 TARGETS = practical1 practical2 practical3 practical4 zombie zombie_fixed
5
6 all: $(TARGETS)
7
8 practical1: practical1.c
9     $(CC) $(CFLAGS) practical1.c -o practical1
10
11 practical2: practical2.c
12     $(CC) $(CFLAGS) practical2.c -o practical2
13
14 practical3: practical3.c
15     $(CC) $(CFLAGS) practical3.c -o practical3
16
17 practical4: practical4.c
18     $(CC) $(CFLAGS) practical4.c -o practical4
19
20 zombie: zombie.c
21     $(CC) $(CFLAGS) zombie.c -o zombie
22
23 zombie_fixed: zombie_fixed.c
24     $(CC) $(CFLAGS) zombie_fixed.c -o zombie_fixed
25
26 clean:
27     rm -f $(TARGETS)
```

Bracket match found on line: 27

Makefile ▼ Tab Width: 8 ▼ Ln 27, Col 25 INS

Skill -1 OS

```
anirudh@Anirudh:~$ cd ~/ossp_practical
anirudh@Anirudh:~/ossp_practical$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
anirudh@Anirudh:~/ossp_practical$ git --version
git version 2.43.0
anirudh@Anirudh:~/ossp_practical$ ls -la
total 156
drwxr-xr-x  2 anirudh anirudh  4096 Aug 12 09:03 .
drwxr-x---  8 anirudh anirudh  4096 Aug 12 06:54 ..
-rw-r--r--  1 anirudh anirudh   34 Aug 12 07:03 destination.txt
-rwxr-xr-x  1 anirudh anirudh 16480 Aug 12 06:56 practical1
-rw-r--r--  1 anirudh anirudh  1014 Aug 12 06:55 practical1.c
-rwxr-xr-x  1 anirudh anirudh 16224 Aug 12 07:03 practical2
-rw-r--r--  1 anirudh anirudh  1185 Aug 12 07:02 practical2.c
-rwxr-xr-x  1 anirudh anirudh 16264 Aug 12 08:28 practical3
-rw-r--r--  1 anirudh anirudh  1054 Aug 12 08:28 practical3.c
-rwxr-xr-x  1 anirudh anirudh 16400 Aug 12 08:55 practical4
-rw-r--r--  1 anirudh anirudh  2139 Aug 12 08:55 practical4.c
-rw-r--r--  1 anirudh anirudh    11 Aug 12 07:08 sample.txt
-rw-r--r--  1 anirudh anirudh    34 Aug 12 07:03 source.txt
-rw-r--r--  1 anirudh anirudh  8102 Aug 12 07:10 trace.txt
-rwxr-xr-x  1 anirudh anirudh 16216 Aug 12 08:58 zombie
-rw-r--r--  1 anirudh anirudh   736 Aug 12 08:58 zombie.c
-rwxr-xr-x  1 anirudh anirudh 16272 Aug 12 09:03 zombie_fixed
-rw-r--r--  1 anirudh anirudh   953 Aug 12 09:03 zombie_fixed.c
anirudh@Anirudh:~/ossp_practical$ gedit Makefile
```

```
anirudh@Anirudh:~/ossp_practical$ gedit Makefile
anirudh@Anirudh:~/ossp_practical$ make
make: Nothing to be done for 'all'.
anirudh@Anirudh:~/ossp_practical$ make -n
make: Nothing to be done for 'all'.
anirudh@Anirudh:~/ossp_practical$ make -B
gcc -Wall -Wextra practical1.c -o practical1
gcc -Wall -Wextra practical2.c -o practical2
gcc -Wall -Wextra practical3.c -o practical3
gcc -Wall -Wextra practical4.c -o practical4
gcc -Wall -Wextra zombie.c -o zombie
gcc -Wall -Wextra zombie_fixed.c -o zombie_fixed
anirudh@Anirudh:~/ossp_practical$ ls -l Makefile
-rw-r--r-- 1 anirudh anirudh 562 Aug 13 08:14 Makefile
anirudh@Anirudh:~/ossp_practical$ |
```

Skill -1 OS

```
Open  skill1_process.c  Save  x
~/ossp_practical

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 int main()
8 {
9     pid_t pid;
10
11     printf("Parent process started\n");
12     printf("Parent PID: %d\n", getpid());
13
14     pid = fork();
15
16     if (pid < 0)
17     {
18         perror("fork failed");
19         return 1;
20     }
21
22     if (pid == 0)
23     {
24         /* Child process */
25         printf("\nChild process created\n");
26         printf("Child PID : %d\n", getpid());
27         printf("Child PPID: %d\n", getppid());
28
29         printf("Child executing 'ls' using exec()\n");
30
31         execlp("ls", "ls", NULL);
32
33         perror("exec failed");
34         exit(1);
35     }
36     else
37     {
38         /* Parent process */
39         printf("\nParent process\n");
40         printf("Parent PID: %d\n", getpid());
41         printf("Child PID : %d\n", pid);
42
43         wait(NULL);
44
45         printf("\nChild process completed.\n");
46         printf("Parent process completed.\n");
47     }
48
49     return 0;
50 }
```

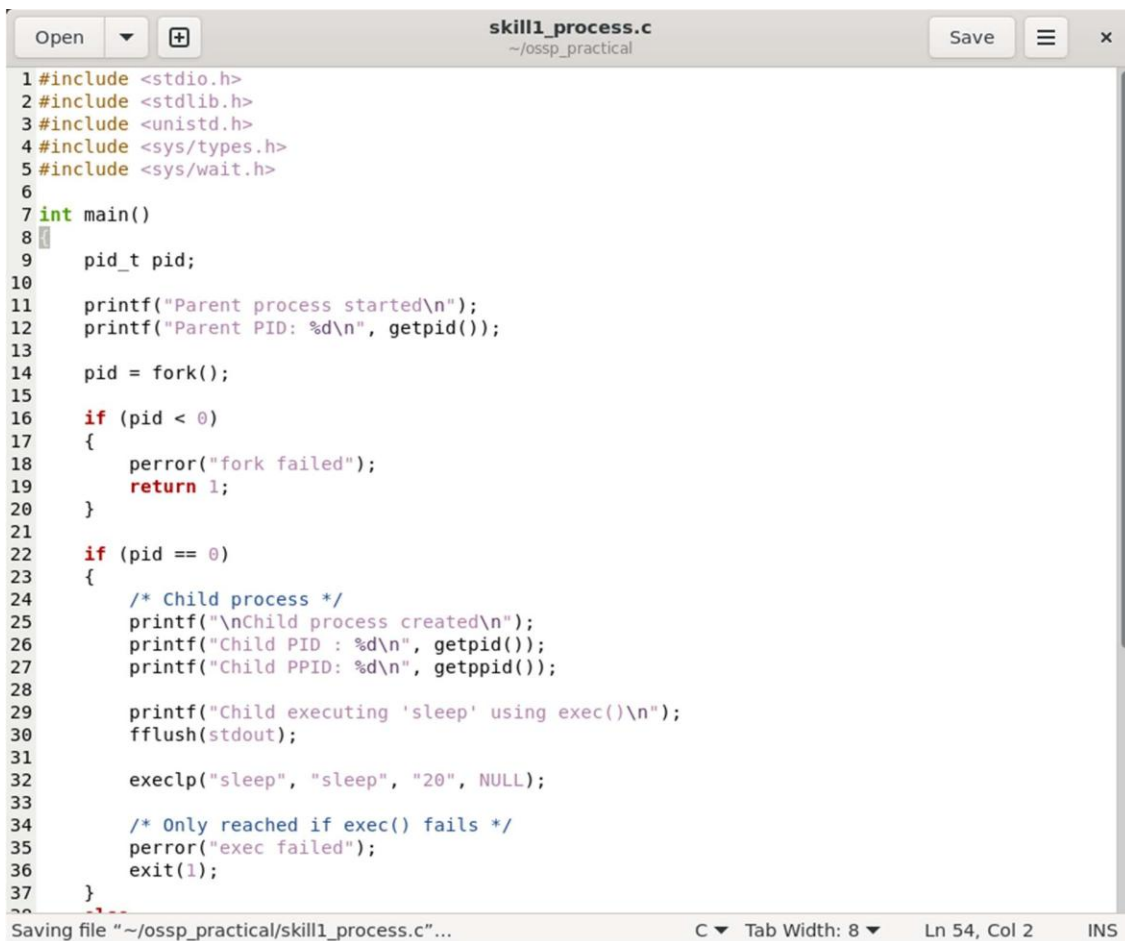
Saving file "~/ossp_practical/skill1_process.c"...

C Tab Width: 8 Ln 50, Col 2 INS

Skill -1 OS

```
anirudh@Anirudh:~/ossp_practical$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-08-13 08:20:44 IST; 2min 17s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 1224 (apache2)
      Tasks: 55 (limit: 4603)
     Memory: 22.1M
        CPU: 231ms
    CGroup: /system.slice/apache2.service
            └─1224 /usr/sbin/apache2 -k start
              └─1225 /usr/sbin/apache2 -k start
                └─1226 /usr/sbin/apache2 -k start
                  └─1227 /usr/sbin/apache2 -k start
                    └─1228 /usr/sbin/apache2 -k start

Aug 13 08:20:44 Anirudh systemd[1]: Starting The Apache HTTP Server...
Aug 13 08:20:44 Anirudh apachectl[1220]: AH00558: apache2: Could not reliably determine
the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName'
Aug 13 08:20:44 Anirudh systemd[1]: Started The Apache HTTP Server.
Aug 13 08:20:44 Anirudh systemd[1]: Reloading The Apache HTTP Server.
Aug 13 08:20:44 Anirudh systemd[1]: Reloaded The Apache HTTP Server.
anirudh@Anirudh:~/ossp_practical$
```



```
skill1_process.c
~/ossp_practical

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 int main()
8 {
9     pid_t pid;
10
11     printf("Parent process started\n");
12     printf("Parent PID: %d\n", getpid());
13
14     pid = fork();
15
16     if (pid < 0)
17     {
18         perror("fork failed");
19         return 1;
20     }
21
22     if (pid == 0)
23     {
24         /* Child process */
25         printf("\nChild process created\n");
26         printf("Child PID : %d\n", getpid());
27         printf("Child PPID: %d\n", getppid());
28
29         printf("Child executing 'sleep' using exec()\n");
30         fflush(stdout);
31
32         execlp("sleep", "sleep", "20", NULL);
33
34         /* Only reached if exec() fails */
35         perror("exec failed");
36         exit(1);
37     }
38 }
```

Saving file "~/ossp_practical/skill1_process.c"...

C Tab Width: 8 Ln 54, Col 2 INS

Skill -1 OS

```
29     printf("Child executing 'sleep' using exec()\n");
30     fflush(stdout);
31
32     execlp("sleep", "sleep", "20", NULL);
33
34     /* Only reached if exec() fails */
35     perror("exec failed");
36     exit(1);
37 }
38 else
39 {
40     /* Parent process */
41     printf("\nParent process\n");
42     printf("Parent PID: %d\n", getpid());
43     printf("Child PID : %d\n", pid);
44
45     printf("Parent waiting for child...\n");
46
47     wait(NULL);
48
49     printf("\nChild process completed.\n");
50     printf("Parent process completed.\n");
51 }
52
53 return 0;
54 }
```

C ▾ Tab Width: 8 ▾ Ln 54, Col 2 INS

```
anirudh@Anirudh:~/ossp_practical$ gcc -Wall -Wextra skilll_process.c -o skilll_process
anirudh@Anirudh:~/ossp_practical$ ./skilll_process
Parent process started
Parent PID: 16136

Parent process
Parent PID: 16136
Child PID : 16137
Parent waiting for child...

Child process created
Child PID : 16137
Child PPID: 16136
Child executing 'sleep' using exec()

Child process completed.
Parent process completed.
anirudh@Anirudh:~/ossp_practical$ strace -f -e trace=process ./skilll_process
execve("./skilll_process", [ "./skilll_process" ], 0x7ffe88ac5678 /* 26 vars */) = 0
[pid 16136]
Parent process started
Parent PID: 16136
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD,
child_tidptr=0x7f8e35f4a2a10) = 16137
[pid 16137]
Parent process
Parent PID: 16136
Child PID : 16137
Child PPID: 16136
Child executing 'sleep' using exec()
[pid 16137] +++ exited with 0 +++
[pid 16136] --- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=16137,
si_uid=1000, si_status=0, si_etime=0, si_stime=0} ---
Child process completed.
Parent process completed.
[pid 16136] +++ exited with 0 +++
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=16136, si_uid=1000,
si_status=0, si_etime=0, si_stime=0} ---
anirudh@Anirudh:~/ossp_practical$
```


Skill -1 OS

```
anirudh@Anirudh:~/ossp_practical$ strace -f -e trace=process ./skill1_process
execve("./skill1_process", ["/./skill1_process"], 0x7ffc2c41c538 /* 26 vars */) = 0
Parent process started
Parent PID: 16168
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, trace: Process 16169 attached
, child_tidptr=0x7f9243a1ca10) = 16169

Parent process
Child process created
Parent PID: 16168
Child PID : 16169
Child PID : 16169
Parent waiting for child...
[pid 16168] wait4(-1, Child PPID: 16168
Child executing 'sleep' using exec(C)

[pid 16169] <unfinished ...>
[pid 16169] execve("/usr/local/sbin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = -1 ENOENT (No such file or
directory)
[pid 16169] execve("/usr/local/bin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = -1 ENOENT (No such file or d
irectory)
[pid 16169] execve("/usr/sbin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = -1 ENOENT (No such file or direct
ory)
[pid 16169] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = 0
[pid 16169] exit_group(0)
[pid 16169] +++ exited with 0 +++
<... wait4 resumed>NULL, 0, NULL) = 16169
anirudh@Anirudh:~/ossp_practical$
```

```
ory)
[pid 16169] execve("/usr/bin/sleep", ["sleep", "20"], 0x7fffd77094058 /* 26 vars */) = 0
[pid 16169] exit_group(0) = ?
[pid 16169] +++ exited with 0 +++
<... wait4 resumed>NULL, 0, NULL) = 16169
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=16169, si_uid=1000, si_status=0,
si_etime=0, si_stime=0} ---

Child process completed.
Parent process completed.
anirudh@Anirudh:~/ossp_practical$ strace -f -e trace=process ./skill1_process 2>&1 | grep -E 'clone|fork|execve'
execve("./skill1_process", ["/./skill1_process"], 0x7ffdc9e1daa38 /* 26 vars */) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, trace: Process 16175 attached
, child_tidptr=0x7febd8a8aa10) = 16175
[pid 16175] execve("/usr/local/sbin/sleep", ["sleep", "20"], 0x7ffec6c2e218 /* 26 vars */) = -1 ENOENT (No such file or
directory)
[pid 16175] execve("/usr/local/bin/sleep", ["sleep", "20"], 0x7ffec6c2e218 /* 26 vars */) = -1 ENOENT (No such file or d
irectory)
[pid 16175] execve("/usr/sbin/sleep", ["sleep", "20"], 0x7ffec6c2e218 /* 26 vars */) = -1 ENOENT (No such file or direct
ory)
[pid 16175] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffec6c2e218 /* 26 vars */) = 0
[pid 16175] exit_group(0) = ?
[pid 16175] +++ exited with 0 +++
<... wait4 resumed>NULL, 0, NULL) = 16175
Parent waiting for child...
exit_group(0) = ?
+++ exited with 0 +++
anirudh@Anirudh:~/ossp_practical$
```

Skill -1 OS

```
anirudh@Anirudh:~$ pstree -p
systemd(1)
├─agetty(234)
├─cron(173)
├─dbus-daemon(174)
├─init-systemd(ub(2))
│   ├──SessionLeader(358)─Relay(360)(359)─bash(360)─skill1_process(16136)─sleep(16137)
│   ├──SessionLeader(16138)─Relay(16143)(16139)─bash(16143)─pstree(16156)
│   ├──init(7)─init(8)
│   ├──login(361)─bash(430)
│   └─{init-systemd(ub)(9)}
├─polkitd(11931)
│   ├──polkitd(11933)
│   ├──polkitd(11934)
│   └─polkitd(11935)
├─rsyslogd(12845)
│   ├──rsyslogd(12847)
│   ├──rsyslogd(12848)
│   └─rsyslogd(12849)
├─systemd(405)
│   ├──(sd-pam)(406)
│   ├──at-spi-bus-laun(553)
│   │   ├──dbus-daemon(560)
│   │   ├──at-spi-bus-laun(554)
│   │   ├──at-spi-bus-laun(555)
│   │   ├──at-spi-bus-laun(556)
│   │   ├──at-spi-bus-laun(558)
│   │   ├──at-spi2+registr(589)
│   │   ├──at-spi2+registr(590)
│   │   └─at-spi2+registr(591)
│   ├──dbus-daemon(549)
│   ├──dconf-service(600)
│   │   ├──dconf-service(601)
│   │   ├──dconf-service(602)
│   │   └─dconf-service(603)
│   ├──xdg-desktop-por(561)
│   │   ├──xdg-desktop-por(562)
│   │   ├──xdg-desktop-por(563)
│   │   ├──xdg-desktop-por(564)
│   │   └─xdg-desktop-por(593)
│   ├──xdg-desktop-por(581)
│   │   ├──xdg-desktop-por(582)
│   │   ├──xdg-desktop-por(583)
│   │   ├──xdg-desktop-por(585)
│   │   ├──xdg-desktop-por(586)
│   │   └─xdg-desktop-por(587)
│   ├──xdg-document-po(566)
│   │   ├──fusemount3(577)
│   │   ├──xdg-document-po(567)
│   │   ├──xdg-document-po(568)
│   │   ├──xdg-document-po(569)
│   │   ├──xdg-document-po(576)
│   │   ├──xdg-document-po(578)
│   │   ├──xdg-document-po(579)
│   │   └─xdg-document-po(580)
│   └─xdg-permission-(570)
│       ├──xdg-permission-(571)
│       ├──xdg-permission-(572)
│       └─xdg-permission-(574)
├─systemd-journal(2722)
├─systemd-logind(1195)
├─systemd-resolve(2902)
├─systemd-timesyn(2785)─systemd-timesyn(2786)
├─systemd-udev(2844)
│   ├──(udev-worker){16140}
│   ├──(udev-worker){16141}
│   └─(udev-worker){16142}
├─unattended-upgr(277)─unattended-upgr(322)
└─ssl-pro-service(199)
    ├──ssl-pro-service(268)
    ├──ssl-pro-service(269)
    └─ssl-pro-service(270)
```

```
anirudh@Anirudh:~$ pstree -p
├─xdg-document-po(576)
├─xdg-document-po(578)
├─xdg-document-po(579)
└─xdg-permission-(570)
    ├──xdg-permission-(571)
    ├──xdg-permission-(572)
    └─xdg-permission-(574)

├─systemd-journal(2722)
├─systemd-logind(195)
├─systemd-resolve(2902)
├─systemd-timesync(2785)─systemd-timesyn(2786)
├─systemd-udev(2844)
│   ├──{udev-worker}{16140}
│   ├──{udev-worker}{16141}
│   └─{udev-worker}{16142}
├─unattended-upgr(277)─unattended-upgr(322)
└─ssl-pro-service(199)
    ├──{ssl-pro-service}{268}
    ├──{ssl-pro-service}{269}
    ├──{ssl-pro-service}{270}
    ├──{ssl-pro-service}{271}
    ├──{ssl-pro-service}{272}
    ├──{ssl-pro-service}{276}
    └─{ssl-pro-service}{307}
```

```
anirudh@Anirudh:~$
```

Skill -1 OS

```
anirudh@Anirudh:~/ossp_practical$ gcc -Wall -Wextra skill1_process.c -o skill1_process
skill1_process.c: In function 'main':
skill1_process.c:27:18: error: 'SIGCHLD' undeclared (first use in this function)
   27 |     signal(SIGCHLD, SIG_IGN);
      |             ^~~~~~

skill1_process.c:27:18: note: each undeclared identifier is reported only once
      for each function it appears in
skill1_process.c:27:27: error: 'SIG_IGN' undeclared (first use in this function)
   27 |     signal(SIGCHLD, SIG_IGN);
      |                     ^~~~~~

skill1_process.c:35:9: error: too few arguments to function 'execl'.
   Have (char *,
   char *, char *, char *)
   35 |     execl("/usr/bin/sleep", "sleep", "20");
      |         ^~~~

In file included from skill1_process.c:2:
/usr/include/unistd.h:528:12: note: declared here
   528 | extern int execl (const char *_path, const char *__arg,
      |             ~~~~~
      |             const char *, ...) __THROW __nonnull ((1));
anirudh@Anirudh:~/ossp_practical$
```